

Chapter 2: Methods for Causal Estimation

2.1 Introduction to Causal Estimation

Causal estimation aims to quantify the effect of an exposure or intervention on an outcome by comparing observed outcomes with the outcomes that would have occurred under alternative exposure scenarios. Unlike descriptive or associational analyses, causal inference focuses on estimating counterfactual quantities.

The foundation of modern causal inference is the potential outcomes framework, introduced by Rubin. For each individual, two potential outcomes exist:

- $Y(1)$: outcome that would occur if the individual receives treatment/exposure.
- $Y(0)$: outcome that would occur if the individual does not receive treatment/exposure.

The individual causal effect is: $Y(1) - Y(0)$

However, only one of these outcomes can be observed for each individual, which creates the fundamental problem of causal inference. Therefore, statistical methods are required to estimate the missing counterfactual outcome.

The most commonly estimated causal parameter is the **Average Treatment Effect (ATE)**:

$$ATE = E[Y(1) - Y(0)]$$

2.2 Assumptions for Causal Estimation

All causal estimation methods require assumptions that allow observed data to approximate counterfactual comparisons.

2.2.1 Consistency

Consistency assumes that the observed outcome corresponds to the potential outcome under the treatment actually received.

$$Y = Y(A)$$

This requires that treatment exposure is well-defined and that different versions of the same treatment do not produce different outcomes.

2.2.2 Exchangeability

Exchangeability assumes that treatment groups are comparable after adjustment for measured confounders.

$$Y(a) \perp A \mid L$$

where:

- A represents treatment/exposure.
- L represents measured confounding variables.

This assumption implies that there is no uncontrolled confounding.

2.2.3 Positivity

Positivity requires that every individual has a non-zero probability of receiving each treatment level.

$$P(A=a | L=l) > 0 \quad \forall a, l$$

For example, individuals with a particular combination of characteristics must have the possibility of receiving both treatment and control conditions.

2.3 Regression Adjustment

Concept

Regression adjustment is one of the most commonly used approaches for estimating causal effects in observational studies. The method estimates the relationship between treatment, covariates, and outcome using a regression model.

A general regression model can be written as:

$$Y = \beta_0 + \beta_1 A + \beta_2 L_1 + \beta_3 L_2 + \epsilon \quad \text{where:}$$

- Y represents the outcome.
- A represents treatment/exposure.
- L represents measured confounders.

The coefficient associated with treatment represents the estimated effect conditional on covariates.

Standardization using Regression Models

Rather than directly interpreting regression coefficients, causal inference commonly uses standardization.

The steps are:

1. Estimate the outcome model.
2. Predict outcomes assuming everyone received treatment:
3. Predict outcomes assuming everyone remained untreated:
4. Calculate the average difference:

Advantages

- Simple implementation.
- Familiar statistical framework.

- Efficient when the model is correctly specified.

Limitations

- Requires correct specification of the outcome model.
- Sensitive to incorrect functional forms.
- Does not account for unmeasured confounding.

2.4 Stratification

Concept

Stratification estimates causal effects by dividing the population into groups based on confounding variables and comparing treatment groups within each stratum.

For example, age may be considered a confounder. The population can be divided into:

- <40 years
- 40–60 years
- 60 years

The treatment effect is estimated separately within each stratum.

The overall causal effect is obtained by combining stratum-specific effects:

$$ATE = \sum ATE_s P(s)$$

Advantages

- Transparent approach.
- Easy interpretation.
- Useful for demonstrating confounding.

Limitations

- Requires adequate sample size within strata.
- Difficult with many covariates.
- Limited applicability with continuous variables.

2.5 Matching

Concept

Matching attempts to create comparable treatment and control groups by pairing individuals with similar characteristics.

The goal is to approximate a randomized experiment by balancing observed confounders between groups.

Types of Matching

Exact Matching

Individuals are matched exactly according to selected covariates.

Example:

A treated individual:

- Male
- 50 years old

is matched with a control individual:

- Male
- 50 years old

Propensity Score Matching

Individuals are matched according to their estimated probability of receiving treatment.

The propensity score is defined as:

$$e(X) = P(A=1 | X)$$

where X represents measured covariates.

Common Matching Algorithms

- Nearest-neighbour matching
- Caliper matching
- Mahalanobis distance matching
- Optimal matching

Advantages

- Improves comparability between groups.
- Provides intuitive assessment of covariate balance.

Limitations

- May reduce sample size.
- Sensitive to matching decisions.
- Cannot remove unmeasured confounding.

2.6 Propensity Score

Concept

- The propensity score was introduced by Rosenbaum and Rubin .
- In this chapter, Herman introduces it as the probability that a person receives treatment given their characteristics.
- Instead of modelling, $Y = f(A=1)$, **The propensity score is: $\pi(L) = P(A=1|L)$.**
- It represents the probability of receiving treatment conditional on observed covariates.
- Propensity Scores help as the treatment received depends on the covariates.
- After PS adjustment, people with similar PS are comparable..
- Each person may have different characteristics, but with the same PS the treatment assigned is balanced.

Applications of Propensity Scores

Propensity scores can be used for:

1. Matching: We tend to pair individuals, who are considered similar in terms of their likelihood of receiving treatment. The PS plots, should overlap, Problem is that when matched, we might miss some population, so the matched population is not the targeted population
2. Stratification: we create strata (with quantiles/deciles), and estimate the treatment effect in each strata. But the problem is two people in the same strata may still have differences in PS so even after adjusting, there might be some bias let out in the treatment effect.
If we use continuous PS, then the relationship could be nonlinear, so there would be a problem if we assume the PS to be linear.
3. Standardization: Predict counterfactual outcomes under each treatment and average them over the target population.
4. IP weighting and G estimation also.

Advantages

- Reduces dimensionality.
- Allows balancing of multiple covariates.
- Widely applicable in observational studies

Limitations

- Adjusts only measured confounders.
- Depends on correct model specification.

2.7 Inverse Probability of Treatment Weighting (IPTW)

Concept

Inverse Probability of Treatment Weighting creates a pseudo-population in which treatment assignment is independent of measured confounders.

Each individual receives a weight based on the inverse probability of receiving their observed treatment.

$$w_i = \begin{cases} \frac{1}{P(A=1|L)}, & A = 1 \\ \frac{1}{1-P(A=1|L)}, & A = 0 \end{cases}$$

This can be written as:

Interpretation : Individuals who received an unlikely treatment receive larger weights.

For example:

A patient with severe disease has a probability of receiving treatment of 0.95. Since treatment was expected, the individual receives a smaller weight.

A patient with mild disease has a probability of receiving treatment of 0.10. Since treatment was unexpected, the individual receives a larger weight.

Three assumptions need to still hold, that is exchangeability, positivity and consistency.

Stabilized Weights

To reduce variability, high CI, unstable estimates, stabilized weights may be used:

$$SW_i = \frac{P(A_i)}{P(A_i | L_i)}$$

Advantages

- Creates balanced pseudo-populations.
- Estimates marginal treatment effects.
- Useful for longitudinal studies.
- Less extreme and more stabilized

Limitations

- Sensitive to extreme weights.
- Requires correct treatment model.
- Vulnerable to positivity violations.

2.8 G-Computation

Concept

G-computation, also called the parametric g-formula, estimates causal effects by modelling the outcome distribution and predicting counterfactual outcomes.

The method consists of:

1. Outcome modelling.
2. Prediction under treatment.
3. Prediction under control.
4. Averaging predicted outcomes.

Steps

Outcome model: $Y=f(A,X)$

Example:

Outcome~Treatment+Age+Sex

Predict treated outcomes: $E(Y | A=1,X)$

Predict untreated outcomes: $E(Y | A=0,X)$

Estimate effect: $ATE=E[E(Y | A=1,L = 1)]-E[E(Y | A=0,L = 1)]$, if the treatment is continuous, then this becomes integral.

Instead of just comparing, we ask the question: what would outcome be under treatment and no treatment for the same distribution of confounders?

It is the causal Identification formula, Here we are talking about parametric, as we assume some relation between the variables.

Why standardization fits:

Standardization + parametric outcome model=Parametric g-formula

Advantages

- Flexible.
- Extends to longitudinal exposures.
- Can incorporate complex outcome models.

Limitations

- Depends on correct outcome model specification.

2.9 Targeted Maximum Likelihood Estimation (TMLE)

Concept

Targeted Maximum Likelihood Estimation is a semiparametric estimation method that combines outcome modelling, treatment modelling, and machine learning.

TMLE uses two models:

1. Outcome model:

$$Q(A,X)=E(Y | A,X)$$

Treatment model:

$$g(X)=P(A | X)$$

TMLE Procedure

1. Estimate the initial outcome model.
2. Estimate the propensity score.
3. **Construct the clever covariate:**

$$H(A,X)=A \cdot g(X) - (1-A) \cdot (1-g(X))$$

4. Update the outcome estimate.
5. Calculate the causal effect.

Advantages

- **Allows flexible machine learning algorithms.**
- **Provides efficient causal estimates.**
- **Has double robustness properties.**

Limitations

- **Computationally complex.**
- **Requires specialised software.**

2.10 Double Robust Estimation

Concept

Double robust estimators combine:

1. Outcome modelling.
2. Treatment modelling.

The estimator remains consistent if either the outcome model or treatment model is correctly specified.

Augmented Inverse Probability Weighting (AIPW)

The general form is:

$$\hat{\psi} = 1/n \sum [A(Y-Q_1(X)) \cdot g(X) - (1-A)(Y-Q_0(X)) + Q_1(X) - Q_0(X) \cdot (1-g(X))]$$

where:

- $Q(X)$ represents the outcome model.
- $g(X)$ represents the treatment model.

Interpretation of Double Robustness

Outcome model	Treatment model	Result
Correct	Incorrect	Unbiased estimate
Incorrect	Correct	Unbiased estimate
Incorrect	Incorrect	Biased estimate

Advantages

- Provides protection against model misspecification.
- Forms the basis of modern causal machine learning methods.
- Used in AIPTW and TMLE.

2.11 Summary of Causal Estimation Methods

Method	Main Principle	Main Requirement
Regression adjustment	Models outcome conditional on covariates	Correct outcome model
Stratification	Estimates effects within covariate groups	Few confounders
Matching	Creates comparable treatment groups	Good matches
Propensity scores	Estimates probability of treatment	Correct treatment model

IPTW	Creates pseudo-population through weighting	Correct propensity model
G-computation	Predicts counterfactual outcomes	Correct outcome model
TMLE	Combines machine learning and causal models	Outcome and treatment models
Double robust estimation	Combines treatment and outcome models	At least one correct model

Ref:

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2. Gruber S, Van Der Laan M. tmle: an R package for targeted maximum likelihood estimation. Journal of Statistical Software. 2012 Nov 16;51:1-35.